RetiGAN: A Hybrid Image Enhancement Method for Medical Images

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***Abstract*—Image enhancement can improve image quality and visual effects well, and there are many outstanding applications in natural images. Medical image enhancement is relatively difficult due to its particularity, such as lack of unified standards, scarce annotation, small data, high accuracy requirements, complex patterns, insufficient image contrast, fuzzy details, and large noise interference. Due to the large gap between natural and medical images, there is a lack of effective universal methods for both types of data. This paper proposes a hybrid model based on the Retinex theory and adversarial generative network to implement the RetiGAN model. Firstly, the image is processed by Gaussian filtering, inverse number transformation and image stretching, and then the data is amplified by the generative adversarial network (GAN) to realize data enhancement. In terms of experiments, nearly a thousand X-ray data based on two public and one private medical image datasets were compared with baseline, the most common image enhancement method, and adequate comparative experiments were conducted with mainstream methods such as LL-Net, Zero-DCE, PELE, etc. Experiments show that the method not only improves the image quality indexes such as PSNR, LPIPS and SSIM but also has remarkable visualization effects and enhancement effects in texture details, which shows that our method has great potential and application space in practical clinical diagnosis.**

Keywords-component; Image enhancement; Medical image enhancement; Hribrid retinex-based method;

I. INTRODUCTION

The quality of the image data is critical, especially in medical imaging. Data with rich details or texture information can often provide doctors with a more accurate and effective basis for diagnosis. However, although there have been many related studies in the field of natural images, many classic works of image enhancement have been produced. However, medical image enhancement has been poor due to its particularity, directly affecting the downstream diagnostic accuracy [1]-[4]. The role of image quality can be seen in Figure 1.

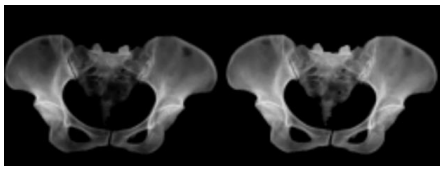

Figure 1. One set of pelvic X-rays (PXR) before and after enhancement shows a significant improvement in details, which helped with follow-up treatment.

In clinical diagnostics, landmark detection for medical images is crucial, particularly for computer-aided diagnosis systems as shown in Figure 2. There has been a surge in the employment of deep learning techniques for landmark detection [1]-[5]. However, image quality directly impacts the accuracy of landmark detection. The unclear image will directly lead to inaccurate positioning of key points and difficulties in model fitting, thus affecting subsequent diagnosis. Therefore, exploring a widely applicable image enhancement method in medical images is urgent.

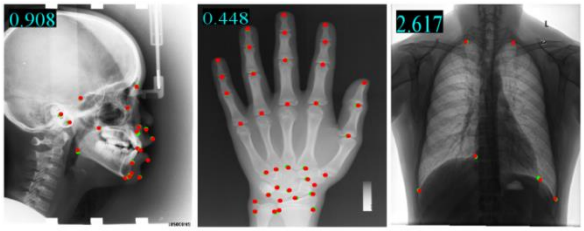

Figure 2. Detection diagram of the key points of the head, hand and chest is very important for the diagnosis and treatment of some diseases such as fractures and pneumonia lesions.

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We think of generative adversarial networks (GANs) [7], which is a generative model that has good generative power and generative effect [6]-[9]. Two neural networks are used against each other to learn data distribution. After the training is complete, the generator can generate realistic virtual images. These virtual images can be used for image generation tasks such as style transfer, image restoration, etc. At the same time, generators can also be used for data enhancement by generating virtual samples to augment the original dataset.

However, although GANs have been revolutionary in their ability to generate high-quality, realistic images by effectively learning the distribution of a given dataset, their focus on the image level means they might not capture all the feature-level variations that can be essential for a model to understand and generalize from the data it is trained on. In other words, GANs may not capture the finer, feature-level details that could be crucial for training more nuanced and robust models. We devised the retinex method [10]-[12] to use in combination with GAN. This is a classical image enhancement algorithm that can improve the image's visual effect through multi-scale Gaussian blur processing and color recovery operations.

In this paper, we proposed a well-designed hybrid model, cleverly integrated retinex algorithm and GAN network, and carefully selected three datasets, which not only achieved a baseline effect over the comparison in image quality but also took landmark detection as a practical clinical task which achieved significant results in multiple indicators such as MRE, SDR.

Our contributions are as follows:

• To the best of our knowledge, we are the first to think of the results of retinex and GAN methods to achieve image enhancement, using a multi-stage approach to achieve good results.

• We not only achieved the best results beyond the baseline in common image quality indicators such as SSIM, PSNR, LPIPS, etc. but also took landmark detection, which is commonly used in clinical practice, as a downstream task and significantly improved in SDR, MRE and other indicators. It has proved that our method has a wide range of potential applications in practical clinical settings.

II. Related Work

A. Image Enhancement

Image enhancement is a useful process in the process of landmark detection. Many methods are used for enhancement, among which making deep learning models is the most common method. In 2004, WANG Xingwei et al. used brightness histogram equalization to enhance colourful medical images. This algorithm made images clearer in detailed places and more efficient than the traditional histogram equalization method [15]. Yue Yongying from Nanjing University of Posts and Telecommunications proposed combining generative adversarial network (GAN) and semi-supervised learning to enhance medical image. It first does supervised learning based on a small number of tagged data and then does unsupervised learning with non-tagged data, which solves the problem of lacking data [16].

B. Retinex Network

The Retinex theory is effective in image enhancement. It divides the figure into the effect of incident light and reflects light, and behaves well in figure enhancement, especially in images with uneven lighting. CHEN Shen and ZHANG JianZhou used the mean template for wave filtering, used the multi-scale Retinex model to enhance the picture, and laid a good foundation for landmark detection [17]. ZHOU Kaiting and ZHENG Lixin used the Retinex model to enhance uneven lighting images. They compared this model with three other traditional enhancement methods and got enhancement in contrast and clarity [18]. The Retinex model is of great use in medical images. I used Retinex to process the low-quality image made by the CMOS or CCD endoscope, allowing real-time processing with low computational complexity. However, it can only unshade the dark parts of the image. Retinex is useful in medical images, but many problems remain to be solved.

C. Landmark Detection

Landmark detection is an important technique in medical scenes. However, because of the diversity of the human body, accurately locating landmarks is quite challenging. Early in 2003, Margrit Betke et al. used an attenuation-based template matching approach based on the original ICP algorithm and improved its time complexity [19]. They used the algorithm for lung CT images and got the correct result for 56 out of 58 pictures. With the development of deep learning, more AI is used in this field. In 2017, Simon Andermatt et al. built a network with three MD-GRU layers, a full connection layer and a function activation layer, using 1218 groups of images for training, with a mean localization error of 1.7 mm [20]. Yu-ChengYeh et al. used the deep leaning method, with 2900 consecutive whole-spine lateral images, to do landmark detection of the spine and used the technique in the diagnosis of spinal disease [21].

III. Method

A. Introduction

This section introduces a novel adaptive approach for medical image segmentation, leveraging the framework of GANs and the retinex algorithm. Based on retinex theory, which is outstanding in the field of natural images, we first use a multi-scale adaptive gain retinex (MSRCR) network [11] to enhance the texture details of the image and then use an improved GAN network to enhance the image further. Distinct from other research endeavours employing GANs, the core of this method lies in harnessing the discriminator's acute proficiency in feature recognition within images. This approach places a diminished emphasis on the utility of the generative capabilities of GANs, focusing instead on the discriminative aspects to achieve efficient image segmentation. The overall model of our approach is shown in Figure 3, and we will explain the process in detail in Section B.

B. Model design

(1) Our method can be divided into two stages： Retinex and the preprocessing module serve as the first part of the image enhancement, and the second part is a GAN for two-stage data enhancement, as shown in Figure 3.

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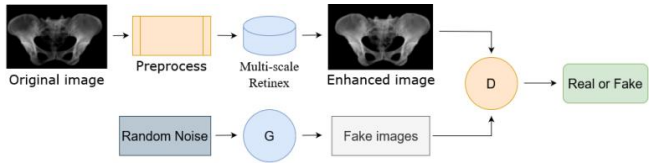

Figure 3. The overall model diagram includes a retinex network and a GAN.

(2) In the initial phase of our methodology, we designed a pre-module based on the retinex algorithm. The data preprocessing includes common operations, including channel number unification, image size adjustment, etc. Because part of our data comes from real clinical practice, a dicom format is adopted, and some of them are full-body long films. To facilitate processing, we used the Yolo network to extract pelvic parts.

(3) The retinex module is the core of image enhancement, and we use a Multi-Scale Retinex algorithm called Multi-scale Retinex (MSR). MSR is developed based on SSR; the advantage is that while maintaining the image's high fidelity and the image dynamic range compression, MSR can also achieve colour enhancement, colour constancy, local dynamic range compression, and global dynamic range compression, and it can also be used for X-ray image enhancement. The most classic is the 3-scale: large, medium and small. The specific steps of this module are as follows: Decompose the original image into multiple scale images.

$$\begin{aligned} I_0 &= G(I_0, \sigma_0) \\ I_1 &= G(I_0, \sigma_1) \\ &\cdots \\ I_n &= G(I_0, \sigma_n) \end{aligned} \tag{1}$$

Where G is a Gaussian filter, σ_i represents the Gaussian kernel scale of the i layer scale, and I_i represents the image of the i layer scale.

(4) Retinex processing was performed on each layer scale image.

$$L_i = \log(I_i) - \log G(I_i, \sigma_g) \tag{2}$$

$$S_i = \log G(I_i, \sigma_s) - \log G(I_i, \sigma_g) \tag{3}$$

$$R_i = \exp(L_i + S_i) \tag{4}$$

Where L_i represents the reflection component of layer i scale, S_i represents the illumination component of layer i scale, and R_i represents the enhanced image of layer i scale.

(5) The enhanced image of each layer is weighted and fused to obtain the final enhanced image.

$$R_f = \frac{\sum_{i=0}^{n} w_i R_i}{\sum_{i=0}^{n} w_i} \tag{5}$$

Where w_i indicates the weight of layer i scale.

Multi-scale Retinex decomposes the original image into multi-scale images, performs Retinex processing on each layer scale's images, and performs weighted fusion to obtain the final enhanced image.

Next, we put the enhanced images after MSR into a GAN to generate data enhancement by generating the same type of data.

We employ the downsampling component of the pre-trained U-Net architecture, applying it to a source domain dataset to construct a discriminator, denoted as D, within the adversarial network framework. Through iterative adversarial training, generator G progressively refines images' synthesis to mirror the target-domain imagery's characteristic attributes more closely.

The overall structure of the domain adaptation method is shown in Figure 3. The function for training these generative adversarial networks is established as:

$$\min \max V(D,G) = E_{x \sim P_{data}(x)}[\log D(x)] + E_{z \sim P_z(x)}[\log(1 - D(G(z)))] \tag{6}$$

Finally, the trained discriminator model in the GAN is used as the down-sampling part of the segmentation model, and the image of the target domain is input into the model for testing. So far, we have enhanced the original data to improve its texture details and extended the data with GAN to achieve the effect of data enhancement.

IV. Experiments and Results

A. Datasets

In order to verify the robustness of the algorithm and improve the clarity of the experimental results, this study mainly uses publicly accessible image data sets. We collected 850 images from three sources. Dataset 1: Open source CGMH-PelvisSeg, provided by CGMHai Lab1, contains 400 high-resolution PXRs. Dataset 2: An internal dataset of the A 150-image open-source dataset provided by [7]. We retrospectively collected 300 images from partner hospitals that complied with Institutional Review Board (IRB) policies. All data is licensed under the CC-BY-NC-SA Creative Commons license, which we will keep intact. To handle the different sizes of these images, we adjusted them to 512*512. We implement enhancement techniques for all the datasets cited in this paper, such as image inversion, rotation, brightness enhancement, and contrast amplification. These methods aim to enlarge the volume and improve the data set's quality to optimize the neural network's training effect. After the enhancement process, the standardization of image sizes is essential to unify training procedures.

B. Experimental Settings

In our experiment, we employ some evaluation metrics: Peak Signal Noise Ratio (PSNR), Learned Perceptual Image Patch Similarity (LPIPS), and Structural Similarity (SSIM). The formulas for calculating these key performance indicators are as follows:

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$$\mathrm{PSNR} = 10 \cdot \log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}}\right) \tag{7}$$

$$\mathrm{d}(x, x_0) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \| w_l \bullet (y_{hw}^l - y_{0hw}^l) \|_2^2 \tag{8}$$

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c1)(2\sigma_{xy} + c2)}{(\mu_x^2 + \mu_y^2 + c1)(\sigma_x^2 + \sigma_y^2 + c2} \tag{9}$$

PSNR is the most common objective measurement method for evaluating images. The larger the value, the less the image distortion. LPIPS learns the reverse mapping of the generated image to the Ground Truth, forces the generator to learn to reconstruct the reverse mapping of the real image from the fake image and prioritizes the perceptual similarity between them. SSIM uses the mean value as an estimate of brightness, standard deviation as an estimate of contrast, and covariance as a measure of structural similarity.

For a baseline comparison, we selected the LL-Net [13], Zero-DCE [14] and PELE. Our baseline approach includes common contrast adjustments, brightness enhancements, and more. For the training process, an Adam optimizer was employed. We use an RTX3090 GPU for training and inference. The batch size is set to 8, the learning rate (LR) is [1e-5, 1e-2], and a cyclic scheduler is used to decrease LR within this interval. Binary Cross Entropy (BCE) loss and Adam optimizer are used. The dataset is randomly divided into 7/1/2 for train/val/test.

C. Experimental Results and Analysis

Table 1 shows that SSIM, LPIPS and PSNR have significantly improved compared with baseline and LL-NET. This proves that our images have a significant effect on quantitative experiments. The SSIM of our method is 0.931, which is more than 20% higher than the 0.679 and 0.793 of the first two methods. The LPIPS also increased significantly to 0.075. The PSNR improvement was relatively small, reaching 22.174, due to the poor quality of the medical data set itself compared to natural images. Zero-DCE is a new image enhancement method, but originally, it was mainly suitable for natural images, especially low-light images, where the effect could be better than PELE. Compared with PELE, our method is not much improved. PELE contains many complex modules and a special image enhancement module, so our method still needs to be significantly improved, but it has improved to some extent. This proves that our approach is highly competitive.

Table 1 quantitative results before and after enhancement, the arrow up behind indicates that the larger the value, the better, and the bold represents the best result.

Method	SSIM↑	LPIPS↓	PSNR (db)↑
Baseline	0.679	0.134	19.071
LL-Net	0.793	0.105	20.796
PELE	0.923	0.083	21.856
Zero-DCE	0.895	0.079	21.977
RetiGAN(ours)	**0.931**	**0.075**	**22.174**

At the same time, we examine the effects of image enhancement from a visual perspective, which is also related to clinical effects. It is clear from Figure 4 that RetiGAN has a strong enhancement effect, greatly improving texture and detail information. Clearer details allow them to more easily identify key points such as CE Angle, acetabular index, H-line, etc., thus more accurately identifying diseases such as fractures. This is of great help to some downstream diagnoses and clinical readings of doctors.

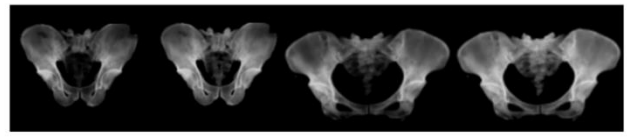

Figure 4. The effects before and after our approach (RetiGAN), data set derived from the pelvis. The first and third columns show two different pre-enhancement effects. The second and fourth columns are the enhanced effects of the previous column, respectively.

V. Conclusions

This paper proposes a method based on retinex theory to enhance images common in medical images, such as low pixels, insufficient contrast, and weak texture details. At the same time, GAN is used to expand the data set to achieve the effect of data enhancement. Experiments show that our method is effective on multiple indicators and data sets and has achieved good results. The visualization effect is also obvious; the image quality is improved, and the downstream diagnosis is significantly beneficial.

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